

Counting edge-disjoint random cliques

Abstract

Choose t copies of K_k uniformly and independently from K_n . When $k \rightarrow \infty$, $k^2 = o(n)$, and $t = \Theta(n^2/k^3)$, the probability that these cliques are pairwise edge-disjoint satisfies

$$\log \zeta(n, k, t) = - \left(\frac{1}{4} + o(1) \right) \frac{t^2 k^4}{n^2}.$$

For the upper bound we estimate the sum of weights of independent sets in an auxiliary graph, using a maximum-degree ordering and a geometric sum. For the lower bound we combine a known locality theorem for random greedy algorithms with a count of bipartite graphs without 4-cycles, then delete a small surplus.

1 Introduction

Two cliques are *edge-disjoint* if they have no common edge. They may share a vertex, but they cannot share two vertices. Choose t cliques $K^1, \dots, K^t$, each with k vertices, uniformly and independently from the complete graph K_n . Write

$$\zeta(n, k, t) = \mathbb{P}(K^1, \dots, K^t \text{ are pairwise edge-disjoint}).$$

Thus t is the number of cliques and k is their order. There are $\binom{n}{k}^t$ equally likely ordered choices.

Acan and Kahn [1, Question 1.4] asked how small this probability is. For $1 \ll k \ll \sqrt{n}$, two random k -cliques share an edge with probability $(1 + o(1))k^4/(2n^2)$. Treating the collisions between different pairs as independent therefore suggests

$$\log \zeta(n, k, t) \sim - \frac{t^2 k^4}{4n^2} \quad (t \rightarrow \infty).$$

Acan and Kahn proved the upper bound in this prediction for $1 \ll t \ll n^2/k^3$ [1, Theorem 1.5(b)]. The next range, $t = \Theta(n^2/k^3)$, is also relevant to packing near-maximum cliques in a random graph; see Griffiths and Mattos [2, Section 1.2].

Problem 1.1 (The original critical-range question). Determine the sharp exponential rate of $\zeta(n, k, t)$ when

$$1 \ll k \ll \sqrt{n}, \quad t = \Theta(n^2/k^3).$$

In particular, for every fixed $D > 0$, prove or disprove the proposed uniform bound

$$\zeta(n, k, t) \leq \exp \left[- \left(\frac{1}{4} - o(1) \right) \frac{t^2 k^4}{n^2} \right] \quad \left(1 \leq t \leq D \frac{n^2}{k^3} \right).$$

The case $k = \Theta(\log n)$ is already of particular interest.

The all- t formulation needs the correction $t^2 \mapsto t(t-1)$, since $\zeta(n, k, 1) = 1$. This does not change the asymptotic statement when $t \rightarrow \infty$. The following theorem gives the critical rate.

Theorem 1.2. *Fix $0 < c \leq D < \infty$. As $k \rightarrow \infty$ and $k^2/n \rightarrow 0$, uniformly for*

$$c \frac{n^2}{k^3} \leq t \leq D \frac{n^2}{k^3},$$

we have

$$\log \zeta(n, k, t) = - \left(\frac{1}{4} + o(1) \right) \frac{t^2 k^4}{n^2}.$$

Corollary 1.3. *For every fixed $D > 0$ there is $\varepsilon_D(n, k) \rightarrow 0$, as $k \rightarrow \infty$ and $k^2/n \rightarrow 0$, such that*

$$\zeta(n, k, t) \leq \exp \left[- \left(\frac{1}{4} - \varepsilon_D(n, k) \right) \frac{t(t-1)k^4}{n^2} \right] \quad \left(1 \leq t \leq D \frac{n^2}{k^3} \right).$$

When $t \rightarrow \infty$, one may replace $t(t-1)$ by t^2 .

The two graphs used in the proof. For the upper bound, we inspect the vertices of K_n one at a time. We put an edge between two clique labels if their cliques have already met. The labels allowed at the next vertex must form an independent set in this graph. A direct weighted count of independent sets gives the upper bound.

For the lower bound, we join each clique label to its vertices, obtaining a bipartite graph. A 4-cycle in this graph means that two cliques have two common vertices. We count bipartite graphs without 4-cycles in which most left degrees are slightly larger than k , then keep t suitable left vertices and k neighbours at each of them. The only external probabilistic input to this part is the locality theorem of Krivelevich, Mészáros, Michaeli and Shikhelman [3]; its use is specified in Section 4. The smaller values of t in Corollary 1.3 are covered by Acan–Kahn’s theorem already quoted above.

All vertices and clique labels are labelled, and all logarithms are natural. We write $x_+ = \max\{x, 0\}$. Constants denoted by C_D may be enlarged from one occurrence to the next, but depend only on D .

2 A weighted count of independent sets

An independent set contains no adjacent pair. Give such a set I weight $z^{|I|}$, where $z > 0$, and write

$$Z_G(z) = \sum_{I \text{ independent in } G} z^{|I|}.$$

Thus the empty set has weight one. The notation $G[U]$ means the graph on U with all edges of G between its vertices, and $e(G[U])$ counts those edges. We estimate the weight sum directly, rather than first counting independent sets of every possible size. The maximum-degree ordering below is the elementary Kleitman–Winston method; see Samotij [4, Section 2]. We include the short weighted calculation because the usual unweighted bound is not the estimate needed here.

Lemma 2.1. *Suppose that G has v vertices and*

$$2e(G[U]) \geq p|U|^2 - \lambda|U| \quad (\emptyset \neq U \subseteq V(G)),$$

where $0 < p < 1$ and $\lambda \geq 0$. For every integer $h \geq 0$ and $z > 0$,

$$Z_G(z) \leq (1-p)^{-h(h+1)/2} \exp \left[z(1-p)^h v + \frac{z\lambda}{p} (1 - (1-p)^h) \right]. \quad (1)$$

Proof. Put $\alpha = 1 - p$ and induct on h , for all graphs satisfying the hypothesis. When $h = 0$, the assertion is just $Z_G(z) \leq (1 + z)^v \leq e^{zv}$. For $h \geq 1$, order the vertices as $v_1, \dots, v_v$ by repeatedly deleting a maximum-degree vertex, breaking ties in a fixed way. If v_r is the first vertex of an independent set in this order, the other vertices belong to the graph H_r obtained by deleting $v_1, \dots, v_r$ and the remaining neighbours of v_r . At this step v_r has degree at least $p(v - r + 1) - \lambda$, so

$$|V(H_r)| \leq \alpha(v - r) + \lambda, \quad Z_G(z) = 1 + z \sum_{r=1}^v Z_{H_r}(z).$$

Every H_r inherits the hypothesis. Apply the induction hypothesis, and put

$$w = z\alpha^h, \quad A = \alpha^{-h(h+1)/2} \exp \left[\frac{z\lambda}{p}(1 - \alpha^h) \right] \geq 1.$$

The preceding identity gives

$$Z_G(z) \leq 1 + Aw \sum_{r=1}^v e^{w(v-r)} \leq 1 + A(e^{wv} - 1) \leq Ae^{wv}.$$

Here the middle inequality is the geometric-sum identity and $e^w - 1 \geq w$. This is exactly (1). The empty graph also satisfies the bound, so all induction steps apply. $\square$

3 The upper bound

Assume throughout this section that $c \leq tk^3/n^2 \leq D$, and put

$$M = tk, \quad d = \frac{M}{n}, \quad h = \lfloor d \rfloor.$$

Here M is the total number of appearances of vertices in the cliques, and d is the average number of cliques containing a vertex. In particular,

$$d \rightarrow \infty, \quad h \geq 1, \quad \frac{hk^2}{n} \leq D, \quad \frac{t^2k^4}{n^2} = \Theta_{c,D}(M).$$

3.1 Which cliques may contain the next vertex?

After r vertices have been inspected, let B_i be the vertices so far assigned to clique label i . We never give a label more than k vertices. Form a graph G_r on $[t]$, joining i and j when $B_i \cap B_j$ is nonempty. The labels assigned to the next vertex must be independent in G_r ; otherwise two cliques would share an old vertex and the new one.

For $r \geq 1$, write

$$\ell_r = \frac{rk}{n}, \quad D_r = \sum_{i=1}^t (\ell_r - |B_i|)_+, \quad p_r = \frac{rk^2}{n^2}, \quad \lambda_r = k + \frac{2kD_r}{n}.$$

For $U \subseteq [t]$, let a_x count the labels in U assigned to an inspected vertex x . Two labels have met at most once. Hence

$$2e(G_r[U]) = \sum_x a_x(a_x - 1), \quad (\ell_r|U| - D_r)_+ \leq \sum_x a_x \leq k|U|.$$

Cauchy–Schwarz now gives

$$2e(G_r[U]) \geq \frac{(\ell_r|U| - D_r)_+^2}{r} - k|U| \geq p_r|U|^2 - \lambda_r|U|. \quad (2)$$

In the last step we used $(z - D)_+^2 \geq z^2 - 2zD$.

3.2 A random construction used only for counting

Choose an ordering π of the n vertices uniformly at random. At step $r + 1$, assign the next vertex to an independent set I of clique labels that have not yet received k vertices. Give this choice weight $z_r^{|I|}$, where

$$p_0 = 0, \quad b_r = -\log(1 - p_r), \quad z_r = \frac{k}{n}(1 - p_r)^{-h}.$$

Choose it with probability $z_r^{|I|}/Z_r$, where Z_r is the sum of weights of allowed choices. The empty set is allowed. This always produces edge-disjoint cliques of orders at most k ; the orders need not all be exactly k .

By Lemma 2.1 and (2), for $r \geq 1$,

$$\log Z_r \leq d + \frac{h(h+1)}{2}b_r + \frac{k\lambda_r}{np_r}((1 - p_r)^{-h} - 1).$$

Since $p_r \leq k^2/n = o(1)$ and $p_r h \leq D$, we have $((1 - p_r)^{-h} - 1)/p_r \leq C_D h$. Hence

$$\log Z_r \leq d + \frac{h(h+1)}{2}b_r + C_D \frac{kh}{n}\lambda_r. \quad (3)$$

At $r = 0$, simply use $Z_0 \leq (1 + k/n)^t \leq e^d$.

Fix a desired ordered packing $\mathcal{C}$ of t cliques of order k . Let m_y be the number of its cliques containing vertex y , so $\sum_y m_y = M$. For a fixed order π , let $P_\pi(\mathcal{C})$ be the probability that the construction produces $\mathcal{C}$. Multiplying its successive choice probabilities gives

$$\begin{aligned} \log P_\pi(\mathcal{C}) &\geq M \log(k/n) - M + h \sum_{r=0}^{n-1} b_r m_{\pi(r+1)} \\ &\quad - \frac{h(h+1)}{2} \sum_{r=0}^{n-1} b_r - C_D \frac{kh}{n} \sum_{r=1}^{n-1} \lambda_r. \end{aligned} \quad (4)$$

All quantities on the right are evaluated along this fixed packing.

3.3 Average first, then count

Average only over the vertex order. At each position, $\mathbb{E}_\pi m_{\pi(r+1)} = d$. Since $b_r \geq p_r$ and $hd - h(h+1)/2 > 0$ for large n , the average of the two terms involving b_r in (4) is at least

$$\left(hd - \frac{h(h+1)}{2} \right) \sum_{r=0}^{n-1} p_r = \frac{k^2(n-1)}{4n} (2hd - h(h+1)) = \frac{d^2 k^2}{4} + o(M).$$

Here $2hd - h(h+1) = d^2 + O(d)$, while $dk^2 = Mk^2/n = o(M)$.

For each fixed clique, its number X of vertices among the first r positions has mean rk/n and variance at most k . One can see the variance bound by writing X as a sum of k indicators: any two have nonpositive covariance under sampling without replacement. Consequently

$$\mathbb{E}(\ell_r - X)_+ \leq \mathbb{E}|X - \mathbb{E}X| \leq \sqrt{k}, \quad \mathbb{E}D_r \leq t\sqrt{k}.$$

It follows that the average of the last error in (4) is at most

$$C_D \frac{kh}{n} (nk + 2kt\sqrt{k}) \leq C_D M \left(\frac{k^2}{n} + \frac{2D}{\sqrt{k}} \right) = o(M).$$

We have therefore proved, for every desired packing $\mathcal{C}$,

$$\mathbb{E}_\pi \log P_\pi(\mathcal{C}) \geq M \log(k/n) - M + \frac{t^2 k^4}{4n^2} - o(M).$$

The arithmetic–geometric mean inequality gives the same lower bound for $\log \mathbb{E}_\pi P_\pi(\mathcal{C})$.

The probabilities $\mathbb{E}_\pi P_\pi(\mathcal{C})$, summed over the desired packings, are at most one. Thus their number is at most

$$\exp \left[M \log(n/k) + M - \frac{t^2 k^4}{4n^2} + o(M) \right].$$

Finally, the elementary factorial estimate $\log(k!) = k \log k - k + O(\log(k+1))$ gives

$$t \log \binom{n}{k} = M \log(n/k) + M + O \left(t \log(k+1) + \frac{Mk}{n} \right) = M \log(n/k) + M + o(M).$$

Dividing the preceding count by $\binom{n}{k}^t$ proves

$$\log \zeta(n, k, t) \leq -\frac{t^2 k^4}{4n^2} + o(M).$$

Every error above is uniform for $c \leq tk^3/n^2 \leq D$.

4 The lower bound

Represent the cliques by a bipartite graph with clique labels on the left and the vertices of K_n on the right. Join a label to the vertices of its clique. Two labels have two common neighbours exactly when there is a 4-cycle, which we also call a square. We therefore need graphs without squares and with all left degrees equal to k .

4.1 The external input

We use Krivelevich–Mészáros–Michaeli–Shikhelman [3, Proposition 1.1 and its proof in Section 2], with the forbidden-set extension in their Sections 1.1.2 and 7. Here the elements are bipartite edges and the forbidden sets are the four edges of a square.

The needed locality statement is as follows. Starting from an edge, find its squares, then repeat from their other edges. If every fixed number of rounds converges in distribution, and the limiting expected number of search paths grows at most exponentially, then the greedy acceptance probability converges to that in the limiting search. We use the rooted form of the cited proof, allowing the initial edge’s inspection time to be prescribed. Applying its finite-neighbourhood coupling at two roots gives joint convergence when their searches have independent limits. Below we verify these hypotheses and derive the counting consequence; the latter is not quoted from [3].

4.2 Counting bipartite graphs without squares

Lemma 4.1. *Suppose $L, R \rightarrow \infty$, $L = o(R^2)$, $R = o(L^2)$, and $m/(LR)^{2/3}$ stays in a fixed compact subinterval of $(0, \infty)$. For each fixed $\rho > 0$, there are at least*

$$\frac{(LR)^m}{m!} \exp \left[-\frac{m^4}{4L^2 R^2} - o(m) \right]$$

labelled bipartite graphs with parts of sizes L, R , exactly m edges, and no 4-cycle, in which at least $(1 - \rho)L$ left vertices have degree at least $(1 - \rho)m/L$.

Proof. Put $N = LR$. Start with the empty graph and repeatedly choose an edge uniformly from those whose addition creates no square. Write Q_j for the number of available edges after j choices. We first check the local calculation needed to apply [3].

Check the squares near an edge. Assign independent uniform numbers in $[0, 1]$ to the N possible edges, and inspect them in order, rejecting an edge if it creates a square. This gives the same process: the first available edge in the remaining random order is uniform, and a rejected edge never becomes available. Fix $x > 0$ and stop inspecting at $p = xN^{-1/3}$. The inspected graph has independent edges of probability p . The rescaled inspection times, conditional on inspection, are independent uniforms in $[0, x]$. Our assumptions give

$$LRp^3 = x^3, \quad Lp, Rp \rightarrow \infty, \quad Lp^2 + Rp^2 + p = o(1). \quad (5)$$

For a specified inspected edge ab , let A be the other neighbours of b , and B the other neighbours of a in the inspected graph. A square through ab corresponds to an edge between A and B . The binomial variance estimate gives $|A| = (1 + o(1))Lp$ and $|B| = (1 + o(1))Rp$ with probability tending to one. Conditional on these lists the cross-edges are independent, so their number has distribution $\text{Bin}(|A||B|, p)$, tending to a Poisson distribution of mean x^3 .

Repeat the search from the three other edges in each square, not recounting the square from which an edge was reached. For clarity, first stop after B_0 edges and squares have been recorded in this search, where B_0 is fixed. A newly found square is called fresh if its two vertices outside the starting edge have not been found before. The probability of finding a nonfresh new square is $O_{B_0}(Lp^2 + Rp^2 + p) = o(1)$: reusing one extra vertex requires two new edges, and reusing both requires at least one. A succession of fresh squares creates no unlisted square, since each adds a path of length three with two new internal vertices along an existing edge.

These estimates also justify the independent Poisson limits in later searches. Outside the bounded set of vertices already found, two neighbour lists on the same side have an expected $O(Lp^2)$ or $O(Rp^2)$ common vertices. Previously tested portions of a newly inspected list have size $O_{B_0}(Lp)$ or $O_{B_0}(Rp)$ and supply $o(1)$ further neighbours on average. The remaining trials are independent and their list sizes still concentrate about Lp and Rp . Thus the cross-edge tests for successive searches are disjoint, except for the already recorded squares and an event of probability $o(1)$. Reusing a list causes no additional error, because its size is already concentrated. Two new squares in the same search have repeated new endpoints with probability $O_{B_0}(Lp^2 + Rp^2) = o(1)$.

Consequently, the limiting search attaches an independent $\text{Poisson}(x^3)$ number of further squares to each searched edge; each such square supplies three new edges. The same argument from two specified edges gives independent limits, even if the two edges share an original endpoint: a square containing both would require two further edges and has probability $O(Lp^2 + Rp^2)$. The temporary bound B_0 can be removed after any fixed number of rounds, since the limiting search then has finite expected size. There are $(3x^3)^r$ outward paths through r squares on average; allowing up to three moves within a square only changes the exponential constant. The locality result therefore applies to acceptance and availability (the latter puts the tested edge after all inspected edges).

Compute one acceptance probability. In this limiting search put

$$u(s) = x \mathbb{P}(\text{the initial edge is accepted by rescaled time } s), \quad 0 \leq s \leq x.$$

Conditional on the initial edge being inspected at time s , one of its squares blocks it exactly when the three other edges have all been accepted earlier. Those decisions lie in separate branches, since

the initial edge is still absent. Each branch has probability $u(s)/x$ of acceptance before s . Averaging over the Poisson number of squares gives

$$u'(s) = e^{-u(s)^3}, \quad u(0) = 0, \quad s = \int_0^{u(s)} e^{w^3} dw. \quad (6)$$

Thus an inspected edge is accepted with limiting probability $u(x)/x$, and an uninspected edge is available with limiting probability $e^{-u(x)^3}$. The solution u does not depend on the chosen cutoff x .

Pass from probabilities to counts. Let $S(x)$ be the number of accepted edges and $Q(x)$ the number available at inspection time $xN^{-1/3}$. For a fixed left vertex a , let $d_a(x)$ be its accepted degree. The one-edge and two-edge limits, followed by Chebyshev's inequality, give

$$\frac{S(x)}{N^{2/3}} \rightarrow u(x), \quad \frac{Q(x)}{N} \rightarrow e^{-u(x)^3}, \quad \frac{d_a(x)}{N^{2/3}/L} \rightarrow u(x) \quad \text{in probability.} \quad (7)$$

Indeed, a specified edge is accepted with probability $(u(x) + o(1))N^{-1/3}$, and two distinct specified edges are both accepted with probability $(u(x)^2 + o(1))N^{-2/3}$. There are only three pair types: a shared left endpoint, a shared right endpoint, or no shared endpoint. The covariance estimates are therefore uniform. Their sums give variances $o(N^{4/3})$ and $o((N^{2/3}/L)^2)$ for $S(x)$ and $d_a(x)$; the diagonal terms are negligible because $N^{2/3}/L = (R^2/L)^{1/3} \rightarrow \infty$. The availability calculation similarly gives $\text{Var } Q(x) = o(N^2)$. Markov's inequality now shows that all but $o(L)$ left degrees satisfy the last estimate in (7), with probability tending to one.

The function u is continuous, increasing and unbounded. Apply (7) at a finite grid of inspection times, with its last limiting accepted count larger than m . Between grid points S and every degree increase, while Q decreases. Refining the grid therefore proves that the process reaches m choices and, with probability tending to one,

$$Q_j = N(e^{-j^3/N^2} + o(1)) \quad (0 \leq j \leq m), \quad (8)$$

with a uniform error. Bracketing the m th choice between two grid points also proves the required degree condition.

Finally, with probability at least $1/2$, both the degree condition and (8) hold with a deterministic error tending to zero. Every specified ordered edge sequence on this event has probability at most

$$\prod_{j=0}^{m-1} Q_j^{-1} \leq N^{-m} \exp \left[\sum_{j=0}^{m-1} \frac{j^3}{N^2} + o(m) \right] = N^{-m} \exp \left[\frac{m^4}{4N^2} + o(m) \right].$$

There are therefore at least $N^m \exp[-m^4/(4N^2) - o(m)]$ such sequences. Each final graph arises from at most $m!$ orders of its edges. Dividing by $m!$ proves the lemma. $\square$

4.3 Delete the surplus and count its cost

We prove the lower bound in Theorem 1.2. Put $M = tk$, fix $0 < \eta < 1/10$, and take

$$T = \lceil (1 + \eta)t \rceil, \quad m = \lceil (1 + \eta)kT \rceil.$$

Apply Lemma 4.1 with parts of sizes T, n and $\rho = \eta/4$. Its assumptions hold because

$$\frac{T}{n^2} = O(k^{-3}) \rightarrow 0, \quad \frac{n}{T^2} = O_{c,D}((k^2/n)^3) \rightarrow 0,$$

and $m/(Tn)^{2/3}$ is bounded above and away from zero. The resulting family $\mathcal{H}$ has

$$|\mathcal{H}| \geq \binom{Tn}{m} \exp \left[-\frac{m^4}{4T^2n^2} - o(M) \right].$$

Here we only weakened $(Tn)^m/m!$ to $\binom{Tn}{m}$. Each graph has at least t left vertices of degree at least k , since $(1-\rho)T > t$ and $(1-\rho)m/T > k$ for large n .

In each graph choose t such vertices and k neighbours at each, using a fixed rule. Delete the remaining left vertices and edges, and relabel the retained left vertices in increasing order by $[t]$. This gives an ordered packing. For each resulting packing, at most

$$\binom{T}{t} \binom{Tn-M}{m-M}$$

original graphs are possible: choose its old left labels, then its other $m-M$ edges. If $\mathcal{P}$ is the set of resulting packings, the identity

$$\frac{\binom{Tn}{m}}{\binom{Tn-M}{m-M}} = \frac{\binom{Tn}{M}}{\binom{M}{M}}$$

therefore gives

$$|\mathcal{P}| \geq \frac{\binom{Tn}{M}}{\binom{T}{t} \binom{m}{M}} \exp \left[-\frac{m^4}{4T^2n^2} - o(M) \right].$$

Moreover $\binom{Tn}{M} \geq \binom{n}{k}^t$: choosing k edges at each of the first t left vertices gives that many distinct M -edge subsets. Since $m-M = O(\eta M)$,

$$\log \binom{T}{t} = O_\eta(t) = o(M), \quad \log \binom{m}{M} = O(\eta \log(1/\eta)M).$$

Divide by $\binom{n}{k}^t$ to conclude that

$$\log \zeta(n, k, t) \geq -(1 + O(\eta) + o(1)) \frac{t^2 k^4}{4n^2} - O(\eta \log(1/\eta)M) - o(M).$$

We used $m^4/(T^2n^2) = (1 + O(\eta) + o(1))t^2k^4/n^2$. Since $t^2k^4/n^2 \geq cM$, first let $n \rightarrow \infty$ and then $\eta \downarrow 0$. This proves the matching lower bound, uniformly in the critical range, and completes Theorem 1.2.

5 The uniform bound and small values of t

To prove Corollary 1.3, first suppose t is fixed. Let E_{ij} be the event that cliques i, j share an edge, and put $p_2 = \mathbb{P}(E_{12})$. If X is their number of common vertices, then

$$\mathbb{E} \binom{X}{2} = \frac{\binom{k}{2}^2}{\binom{n}{2}}, \quad \mathbb{E} \binom{X}{3} = \frac{\binom{k}{3}^2}{\binom{n}{3}} = o \left(\mathbb{E} \binom{X}{2} \right).$$

Since $\binom{x}{2} - 1 \leq 3\binom{x}{3}$ for $x \geq 3$, it follows that $p_2 = (1 + o(1))k^4/(2n^2)$. Any two distinct events E_{ij} are independent, even when they share one label: condition on that clique, whose collision probability with another random clique is the same for every possible vertex set. The first two terms of inclusion–exclusion consequently give

$$\log \zeta(n, k, t) = -\binom{t}{2} p_2 + O_t(p_2^2) = -\left(\frac{1}{4} + o(1)\right) \frac{t(t-1)k^4}{n^2} \quad (t \geq 2 \text{ fixed}).$$

For $t = 1$ the assertion is exact.

For $t \rightarrow \infty$ with $tk^3/n^2 \rightarrow 0$, use Acan and Kahn [1, Theorem 1.5(b)]; when tk^3/n^2 is bounded away from zero, use the upper bound proved here. These cases give a uniform conclusion: any sequence violating it by a fixed positive amount has a subsequence with fixed t , or with $t \rightarrow \infty$ and $tk^3/n^2 \rightarrow 0$, or with tk^3/n^2 tending to a point of $(0, D]$. Each alternative contradicts one of the preceding bounds. Finally, $t(t-1) = (1-o(1))t^2$ when $t \rightarrow \infty$.

References

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